Supplementary Information

A preregistered model forecasts mRNA–protein concordance in unmeasured cancer contexts across three proteomic platforms

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Supplementary Note 1. Data sources and processing

CPTAC. Paired tumor RNA and protein were the CPTAC pan-cancer data freeze v1.2, downloaded on 1 October 2026. RNA files are RSEM coding upper-quartile log2 abundances and protein files are log2 ratios to a reference intensity. A sample entered the paired set when it was listed as a tumor with both an RNA column and a protein column. The P gene set contains the 7,432 genes measured in at least 8 of the 10 cancer types at a tumor measurement rate of at least 0.70 and present in the RNA gene universe. The numbers of tumors with both RNA and protein are given in Supplementary Table S1.

TCGA RPPA. The protein matrix was the PanCanAtlas Xena file TCGA-RPPA-pancan-clean, downloaded on 1 October 2026. A TCPA data level is not recorded for this file. RNA was the PanCanAtlas batch-corrected RNA-seq matrix. Samples were the intersection of the two matrices after duplicate RPPA sample identifiers were dropped. Antibody names containing _p, pS, pT, pY, cleaved or _c were excluded, and an antibody mapped to more than one gene symbol was dropped. A cancer type was evaluated when at least 30 samples had both measurements. The matrix had 7,754 samples and 258 antibodies, and the confirmation file had 4,154 antibody–cancer pairs, of which 4,045 had a defined observed concordance.

ProCan. The protein matrix was ProCan-DepMapSanger_protein_matrix_8498_averaged.txt (figshare file 34411172). RNA was DepMap 24Q4 log2(TPM + 1), joined by Sanger model

identifier. Lineage labels are the Tissue_type field of the ProCan mapping file. The structure table had 8,498 protein columns, and the confirmation file had 12,159 protein–lineage pairs, of which 6,118 had a defined observed concordance.

APOLLO. Protein data were the Genomic Data Commons open supplement APOLLO1_Level3_Gprot_v061418.n87.csv (SHA-256 621c04d7ee91d4a8295891c03513c872d731aeae772744045d3748040715f2f4, 9,060,133 bytes) for 87 patients of the US Lung Cancer Biospecimen Resource Network, 2012–2018, with resected primary tumors1. Protein values are log2 TMT ratios to a common reference in channel m/z 126. Quantification used the median of unique peptide-spectrum matches, with at least two unique matches. Proteins quantified in at least 50% of samples had missing values filled by k-nearest neighbors, and proteins quantified in fewer than 50% were excluded (ProteomeXchange PXD036025). RNA files were processed with MapSplice 2.2.2, hg38, HTSeq 0.9.1 and GENCODE v28 basic annotation; TPM used the mean transcript length, with the per-million factor computed over protein-coding genes. We used log2(TPM + 1) without restricting the matrix to coding genes. All 87 samples were kept and no further quality filter was applied.

The processed RNA matrix data/processed/apollo_luad/rna_log2tpm.parquet has SHA-256 24b4aeb2a9bced86a53645a28879f18125f69f24a9e9a443df1ebfd26c3cce96 (20,714,409 bytes), and the sample map config/apollo_sample_map.tsv (87 one-to-one identifiers) has SHA-256 4b9045ad3f9faee678cc85e8769391da939bdf9675d1e72f70476b8064cdb6ef. The RNA files had 56,832 symbol rows, no duplicate symbols and no Ensembl identifiers. Of the 19,211 genes used as RNA inputs of the anchor-residual predictor, 1,287 were absent from the APOLLO RNA matrix and were given z = 0. Per-sample TPM sums ranged from 1.87 × 10^6

to 2.90 × 10^6 (median 2.18 × 10^6) because non-coding genes remain in the file; the sums were not used as a filter.

Protein rows whose symbol contained ;, | or , were dropped, and all rows of a symbol that occupied more than one row were dropped. From 7,631 rows, no multi-symbol rows and 25 rows of 9 duplicated symbols (CUX1, HLA-A, HLA-B, HLA-C, HLA-DRB1, MOCS2, NA, RABGAP1L and TMPO) were removed, leaving 7,606 symbols. The upper bounds of units before finite-value filters were 7,148 for P1, 7,318 for P2 and 6,323 for Q1 and Q2. Of the 7,318 P2 genes, 120 had no RNA row, so their observed concordance was undefined; with two further genes without a defined concordance, 7,196 genes entered P2. Public identifiers of APOLLO and CPTAC lung adenocarcinoma samples did not overlap; enrollment of the same patient under different identifiers cannot be excluded.

Other data. CCLE lung (74 cell lines; 6,911 genes with an own-mRNA correlation) and skin (33 cell lines; 5,947 genes) lineages were used during development to compare the anchor-residual predictor with the gene's own mRNA in cell lines. File names, sources and SHA-256 hashes of all downloaded files are in Supplementary Table S9.

Supplementary Note 2. Forecast model and reproducibility of frozen predictions

Features. F1s uses nine features. F1, fitted in the preceding stage, uses the five correlation summaries, the query RNA mean and standard deviation and the percentile of that standard deviation, the mean of the training RNA means, and the training protein measurement rate.

Feature	Definition	Model
rho_mean	Mean of the gene's defined CPTAC correlations in the training cancers	F1, F1s
rho_sd	Standard deviation of those correlations (divisor n; 0 if fewer than two)	F1, F1s
rho_min	Minimum of those correlations	F1, F1s
rho_max	Maximum of those correlations	F1, F1s
n_defined_cancers	Number of training cancers with a defined correlation	F1, F1s
train_protein_rate_mean	Mean fraction of paired tumors in which the protein was measured	F1, F1s

Feature	Definition	Model
holdout_rna_sd_percentile	Percentile of the gene's RNA standard deviation in the query context	F1, F1s
train_rna_mean	Mean of the gene's per-cancer RNA means in the training cancers	F1
holdout_rna_mean	RNA mean in the query context	F1
holdout_rna_sd	RNA standard deviation in the query context	F1
train_rna_mean_percentile_mean	Mean, across training cancers, of the percentile of the gene's RNA mean	F1s
holdout_rna_mean_percentile	Percentile of the gene's RNA mean in the query context	F1s

Model. LightGBM used 300 trees, learning rate 0.05, 31 leaves, a minimum of 20 samples per leaf, row subsample 0.8 at every iteration, column subsample 0.8 and random state 20261001, with all available threads. Missing feature values were replaced by the column median of the training matrix. The 80% and 90% interval half-widths were 0.202 and 0.259, the conformal quantiles of absolute residuals on the stacked leave-one-cancer-out rows.

Reproducibility. For public release, the forecast model was refitted with the frozen data and parameters, because the fitted model objects had not been stored. The model had been fitted twice with identical training rows, the 72,880 rows of results/f1s_predictions.tsv (SHA-256 3ab5b9378020ef2e3710ba70e6e817416d9a20e055ca8c8086c64b02071334d6) that are in the P gene set with a finite observed correlation, code and parameters: at commit 224779714bffd58b02059f51de48d92d1324a24f before the RPPA and ProCan predictions were frozen, and at commit ce7949a323791175ee4d60c0b7d79d8e8c5bda02 before the APOLLO predictions were frozen. The released model reproduced the frozen APOLLO predictions exactly (maximum absolute difference 2.4 × 10^−16 over 7,318 genes) but not the RPPA and ProCan predictions (0.031 and 0.044; Spearman correlations 0.9991 and 0.9992). The input features were identical (maximum absolute difference 1.4 × 10^−14). Refits with 1, 16 and 72 threads and with automatic, row-wise and column-wise histogram construction all produced the same predictions, and every setting was deterministic across repeats, so neither thread count nor histogram construction explains the difference. The script that wrote the

frozen predictions is byte-identical to the released one, the training rows come from the same file, and LightGBM 4.7.0 was installed on 1 October 2026 and recorded both before and after the two fits. We could not identify the cause. The preregistered statistics recomputed with the released model differed from the frozen values by −2.9 × 10^−5 on RPPA and −3.3 × 10^−4 on ProCan. All results in the paper use the frozen predictions, which are released. Code that fits a model on a table in memory and later reads the same table from disk should treat the two as different inputs.

Supplementary Note 3. Preregistration history and departures

The RPPA and ProCan tests (H1–H4) were preregistered at commit f7befd8324bcef63d89cebb57de36c0232b5b18f (2 October 2026, 09:57:46 +0900), and their confirmation output was committed at a4764533d08bd09f404f827f7e01726c6a897cff (10:29:53). For APOLLO, the RNA processing was committed at ef3ba3ff12bdae349666b1fc32d783ecfa4f673b (15:48:40), the preregistration at 87967f098ea1434f39c3cd3109d2edeea1c1a0ef (15:50:05), the frozen predictions at ce7949a323791175ee4d60c0b7d79d8e8c5bda02 (16:16:58), the code freeze at 759b3b1fc6f49db8398ba17f9769b6818bd67703 (16:58:18), the unlock at f1c1c6370187456b831abef46078e509014e61fc (16:59:01) and the confirmation output at 57a3a3816bec154ae0df4f38dfdae5155eedc464 (16:59:47). Each confirmation command was run once.

Departures accepted before preregistration and before unlocking: the first power calculation was discarded, and the rank definition was edited before the counted tests on simulated data. Holm's procedure was applied to one family of four hypotheses (H1–H4) and to two families of two hypotheses for APOLLO (P1 and P2, primary; Q1 and Q2, secondary). Ordered thresholds are 0.0125, 0.05/3, 0.025 and 0.05 for H1–H4, and 0.025 then 0.05 within each

APOLLO family. Because the two APOLLO families were controlled separately, the familywise error rate across them was not controlled; the worst case stated in the preregistration is about 0.10. Primary claims rest on the primary family.

Post hoc analysis plans were committed before calculation: the lookup plan at 703c94594efb81f2764ec1ec5c442398ef52b0c9 (13 October 2026, 16:44:38 +0900; an earlier commit, c33e5aaf54d4608b763d4b10c4d752183aed5045 at 16:41:11, did not parse and was corrected before any calculation, without changes to the context map or cascade), with output at 9984b2dc569d3f1a0f6a6b2b8b5a76852f1db86e (16:57:21); the comparison of the CPTAC mean lookup with the cascade lookup at a77b269950af12c1dc061c7cdf8f2fd2caa51775 (18:05:17); and the pooled-lookup plan at 68a32e5974eb0426032c8310d13ec7d716fc27c7 (14 October 2026, 10:38:30 +0900); the split of the APOLLO L1 pairs by the forecast's CPTAC coverage, planned after the pooled-lookup results had been seen, at 773bd579f9c2bb4eb65e1e0f0cf955fbbe71a6eb (15 October 2026, 11:47:50 +0900), with output at 8720727cadc81ea4fcf488ab1ac536614accda8c; and the reproduction analysis of the forecast model at 32d4e7191a140668a27cd716d728e63daa73aa71 (15 October 2026, 14:07:53 +0900), with output at 1c02e3776e87334edf9e2b71690717b958b1eb31.

Commit times were recorded by git on the analysis workstation and were not certified by a third party. The published history fixes the order of commits through their hashes but does not independently prove their times.

Supplementary Note 4. Power analysis and tests on simulated data

The power file used in the APOLLO preregistration was the second run. All nine scenarios had a rejection rate of 1.0 and no Wilcoxon errors. The P1 source had n = 7,418, mean 0.0969 and median 0.0624, with draw size 7,148. The Q1 source had n = 7,403, mean 0.0319 and

draw size 6,323, and the Q2 source n = 7,403, mean 0.112 and draw size 6,323. Fifteen anchor-residual genes lacked leave-one-cancer-out R1 or R2 predictions and were excluded from Q only. The CPTAC lung adenocarcinoma reference for P2 had n = 7,418, a forecast Spearman correlation of 0.845, an RNA-variance baseline correlation of 0.386 and a difference of 0.459.

On planted data, the primary tests rejected: P1 had n = 7,029, mean difference 0.0643 and stored P = 0, and P2 had n = 7,196, a forecast correlation of 0.928 against 0.544 for the baseline (difference 0.384, P = 0.0005). On null data, neither test rejected; the absolute mean difference was 0.000105, and the absolute Spearman correlations were 0.0229 for the forecast and 0.0162 for the baseline. The planted and null data sets were local checks and were not committed.

Supplementary Note 5. Sensitivity analyses

Non-imputed APOLLO matrix (post hoc). The non-imputed protein file (GDC publication page APOLLO-LUAD-2022, file 48ef3b6e-4546-4833-ab18-debc2fa81a3c, SHA-256 71210627045eb233613ce7e3035f44e5aecb2d24c7da5685dcb551bb0bb4f549, 10,954,279 bytes, downloaded 2 October 2026) was analysed after excluding 13 sample columns outside the sample map, blank gene-symbol rows and duplicated symbols. The P1-style mean difference was 0.0847 (n = 7,027; stored P = 0) and the P2-style difference was 0.231 (forecast 0.752 against baseline 0.521; n = 7,194; P = 0.0005). These are not preregistered tests.

Anchor-free baselines on RPPA and ProCan (post hoc). Mean ρ_AR − ρ_R1 was 0.0250 on RPPA (n = 100) and −0.0489 on ProCan (n = 2,399); mean ρ_AR − ρ_R2 was −0.0343 on RPPA and −0.103 on ProCan. These stage-5 summaries have no intervals. Refitting R1 and R2 with the stage-5 function reproduced the stored coefficients (n = 4,998; maximum absolute difference 9.97 × 10^−17).

Pre-specified descriptive summaries of the APOLLO anchor-residual differences. Median 0.0556, first quartile 0.0106, third quartile 0.131, fraction above 0.05 0.529 and fraction below −0.05 0.0414 (n = 7,029). The median was 0.0151 outside the P set and 0.0623 inside it. By the selected anchor weight λ, the median difference was 0 at λ = 0, 0.00779 at 0.25, 0.0261 at 0.5, 0.0520 at 0.75 and 0.0753 at 1.

Similarity of APOLLO to CPTAC. The preregistered descriptive similarity s was 0.925, inside the CPTAC holdout range of 0.883–0.979.

Supplementary Note 6. Lookup definitions and coverage

The CPTAC mean lookup F0 used the CPTAC correlations stored during model development. Among RPPA antibodies, 100 genes were in the P set, 28 outside it had a gene-specific F0 and 6 received the global mean; among ProCan proteins, the counts were 3,052, 819 and 182; and among APOLLO genes, 6,323, 201 and 794. These counts include genes whose observed concordance is undefined.

The cascade lookup L* recomputed CPTAC correlations for all genes quantified in each cohort, which added 9,318 gene–cohort correlations. Among pairs with a finite observed concordance, F0 was within 10^−8 of the global mean while L* was a CPTAC value (levels L1–L3) for 681 APOLLO genes, 62 RPPA pairs (2 antibodies) and 0 ProCan pairs. Among the evaluated pairs with a matched-cancer correlation (L1 pairs), F0 was the global mean while the lookups had a CPTAC value for 600 APOLLO pairs, all for genes outside the P set, and for no RPPA or ProCan pairs; for all 600, the forecast's gene-level features were at their default values. The pooled lookup is level L2 of the cascade, evaluated on L1 pairs. The context map is in Supplementary Table S10, and the numbers of pairs in each cascade level are in Supplementary Table S8. With only two RPPA antibodies, comparisons in level L3 were unstable (1,034 of 2,000 resamples undefined) and are not interpreted; comparisons in level L4, where the lookup is constant, are undefined.

Supplementary Note 7. Antibody–drug conjugate atlas

The atlas has 957 rows, 29 targets in each of 33 TCGA cancer types. Each row gives the frozen forecast and its 80% interval (the forecast plus or minus the stored half-width, not clipped). Every CLDN18 row states that gene-level assays cannot separate isoforms and that the ADC target is CLDN18.2. Column definitions accompany results/stage7/adc_atlas_final.tsv in the release. The 29 targets were fixed before any analysis of them and are not the set of approved conjugate targets: the label search found approved conjugates against seven further targets (TNFRSF8, CD22, CD33, CD79B, CD19, TNFRSF17 and IL3RA), whose forecasts are in the released genome-wide table.

In a descriptive comparison with the Human Protein Atlas2, 24 of the 29 targets had a defined correlation, and the Spearman correlation between that concordance and the mean forecast was 0.0443. The two quantities differ in kind: the Human Protein Atlas compares staining levels between cancers, whereas the forecast is a within-cancer correlation across patients.

Labels were retrieved through the openFDA drug label and Drugs@FDA interfaces on 15 October 2026 with the queries listed in Supplementary Table S7, covering the term “antibody-drug conjugate” and the payload stems of approved and investigational ADC names. For each product approved by 2 October 2026, the target was taken from the mechanism-of-action section of the label; “none found” means that no product in this search acted on the target.

Supplementary tables

Supplementary Table S1. CPTAC cohorts.

Cancer type	Tumors with RNA and protein	Cohort reference
BRCA	121	3
CCRCC	103	4
COAD	96	5
GBM	99	6
HNSCC	108	7
LSCC	108	8
LUAD	110	9
OV	82	10
PDAC	105	11

Cancer type	Tumors with RNA and protein	Cohort reference
UCEC	95	12
Total	1,027	

Supplementary Table S2. Forecast minus CPTAC mean lookup by membership in the P gene set (post hoc).

Model	Genes	RPPA	ProCan	APOLLO
F1s-LightGBM	All	0.0643 (0.0213 to 0.112); 4,045 pairs	0.0317 (0.0195 to 0.0440); 6,118 pairs	0.0276 (0.0206 to 0.0348); 7,196 genes
F1s-LightGBM	Inside P	0.0718 (0.0349 to 0.117); 3,051	0.0208 (0.0106 to 0.0308); 5,547	0.00187 (−0.00323 to 0.00678); 6,221
F1s-LightGBM	Outside P	0.0405 (−0.0912 to 0.204); 994 (34 antibodies)	0.139 (0.0702 to 0.209); 571 (347 proteins)	0.276 (0.220 to 0.328); 975
F1s-ridge	All	0.0573 (0.0266 to 0.0914)	0.0386 (0.0295 to 0.0470)	0.00462 (−0.00238 to 0.0117)
F1s-ridge	Inside P	0.0573 (0.02932 to 0.09418); 3,051	0.02943 (0.02216 to 0.03676); 5,547	0.007169 (0.003163 to 0.01106); 6,221
F1s-ridge	Outside P	0.04224 (-0.0519 to 0.1495); 994 (34 antibodies)	0.08092 (0.02566 to 0.1354); 571 (347 proteins)	0.1577 (0.1048 to 0.2098); 975

Notes: Differences in Spearman correlation with observed concordance (95% interval from 2,000 unit resamples).

Supplementary Table S3. Discrimination of genes with observed concordance of at least 0.5 (post hoc).

	RPPA	ProCan	APOLLO
Pairs (positives)	4,045 (951)	6,118 (1,993)	7,196 (3,188)
AUROC, forecast	0.746 (0.701–0.787)	0.738 (0.723–0.753)	0.868 (0.860–0.876)
AUROC, CPTAC mean lookup	0.696 (0.636–0.748)	0.725 (0.709–0.739)	0.857 (0.848–0.865)
AUROC, RNA-variance baseline	0.677 (0.630–0.723)	0.620 (0.603–0.637)	0.753 (0.742–0.764)
Forecast − CPTAC mean lookup, all pairs	0.0498 (0.0264 to 0.0774)	0.0136 (0.00739 to 0.0195)	0.0112 (0.00674 to 0.0157)
Forecast − cascade lookup, all pairs	0.0457 (0.0200 to 0.0756)	0.0140 (0.00352 to 0.0239)	0.0110 (0.00396 to 0.0180)
Forecast − cascade lookup, matched cancer (L1)	0.00798 (−0.0238 to 0.0399)	0.0173 (0.00245 to 0.0329)	0.0100 (0.00274 to 0.0173)
Forecast − RNA-variance baseline, all pairs	0.0683 (0.0157 to 0.120)	0.119 (0.103 to 0.135)	0.115 (0.104 to 0.125)

Supplementary Table S4. Within-context and within-antibody comparisons (post hoc).

Comparison	Value
Median within-context Spearman correlation, forecast: RPPA (31 cancer types), ProCan (3 lineages), APOLLO	0.439, 0.440, 0.747
Same, CPTAC mean lookup	0.398, 0.404, 0.719
Same, RNA-variance baseline	0.246, 0.135, 0.492

Comparison	Value
RPPA, forecast − CPTAC mean lookup within cancer type	30 of 31 positive; median 0.0600 (0.0498 to 0.0773); sign test P = 3.0 × 10^−8
RPPA, forecast − cascade lookup within cancer type	28 of 31 positive; median 0.0799 (0.0574 to 0.0982); P = 4.6 × 10^−6
RPPA within antibody: median Spearman correlation across cancer types, forecast and RNA-variance baseline (130 antibodies)	0.255 and 0.203; median difference 0.0242 (−0.0161 to 0.0959)

Supplementary Table S5. Pre-specified descriptive metrics of all predictors on APOLLO (7,196 genes).

Predictor	Spearman	AUROC (ρ ≥ 0.5)	Pearson	Mean absolute error	Bias
F1s-LightGBM (forecast)	0.747	0.868	0.750	0.116	−0.0227
F1s-ridge	0.724	0.853	0.730	0.120	−0.0242
F1-LightGBM	0.627	0.808	0.620	0.222	0.189
F1-ridge	0.719	0.852	0.722	0.186	0.155
CPTAC mean lookup (F0)	0.719	0.857	0.726	0.121	−0.0124
RNA-variance baseline	0.492	0.753	n/a	n/a	n/a

Notes: Bias is the mean of observed minus predicted concordance. One hundred and twenty-two genes of the 7,318 P2 genes had no defined observed concordance and were excluded.

Supplementary Table S6. Coverage of 80% intervals by calibration method and number of calibration genes K.

Platform	Method	K	Mean 80% coverage (2.5th–97.5th)	Mean width
F1s-LightGBM, the preregistered forecast				
RPPA	No adjustment	10	0.511 (0.491–0.539)	0.404
RPPA	No adjustment	20	0.511 (0.491–0.539)	0.404
RPPA	No adjustment	50	0.511 (0.491–0.539)	0.404
RPPA	No adjustment	100	0.511 (0.491–0.539)	0.404
RPPA	Shift (δ)	10	0.615 (0.540–0.671)	0.404
RPPA	Shift (δ)	20	0.618 (0.542–0.664)	0.404
RPPA	Shift (δ)	50	0.621 (0.579–0.659)	0.404
RPPA	Shift (δ)	100	0.623 (0.595–0.659)	0.404
RPPA	Shift + quantile	10	0.775 (0.633–0.924)	0.566
RPPA	Shift + quantile	20	0.796 (0.693–0.890)	0.581

Platform	Method	K	Mean 80% coverage (2.5th–97.5th)	Mean width
RPPA	Shift + quantile	50	0.798 (0.714–0.879)	0.579
RPPA	Shift + quantile	100	0.801 (0.734–0.853)	0.582
ProCan	No adjustment	10	0.628 (0.617–0.640)	0.404
ProCan	No adjustment	20	0.628 (0.617–0.640)	0.404
ProCan	No adjustment	50	0.628 (0.617–0.640)	0.404
ProCan	No adjustment	100	0.628 (0.617–0.640)	0.404
ProCan	Shift (δ)	10	0.604 (0.501–0.646)	0.404
ProCan	Shift (δ)	20	0.609 (0.548–0.649)	0.404
ProCan	Shift (δ)	50	0.616 (0.580–0.649)	0.404
ProCan	Shift (δ)	100	0.620 (0.592–0.647)	0.404
ProCan	Shift + quantile	10	0.735 (0.587–0.889)	0.545
ProCan	Shift + quantile	20	0.767 (0.581–0.892)	0.574
ProCan	Shift + quantile	50	0.789 (0.701–0.850)	0.586
ProCan	Shift + quantile	100	0.792 (0.745–0.851)	0.587
APOLLO	No adjustment	10	0.833 (0.824–0.846)	0.404
APOLLO	No adjustment	20	0.833 (0.824–0.846)	0.404
APOLLO	No adjustment	50	0.833 (0.824–0.846)	0.404
APOLLO	No adjustment	100	0.833 (0.824–0.846)	0.404
APOLLO	Shift (δ)	10	0.816 (0.739–0.845)	0.404
APOLLO	Shift (δ)	20	0.824 (0.737–0.843)	0.404
APOLLO	Shift (δ)	50	0.830 (0.810–0.843)	0.404
APOLLO	Shift (δ)	100	0.832 (0.819–0.844)	0.404
APOLLO	Shift + quantile	10	0.719 (0.499–0.947)	0.342
APOLLO	Shift + quantile	20	0.742 (0.537–0.906)	0.342
APOLLO	Shift + quantile	50	0.779 (0.673–0.893)	0.360
APOLLO	Shift + quantile	100	0.791 (0.722–0.861)	0.364
F1s-ridge, not the preregistered predictor				
RPPA	No adjustment	10	0.472 (0.425–0.520)	0.404

Platform	Method	K	Mean 80% coverage (2.5th–97.5th)	Mean width
RPPA	No adjustment	20	0.472 (0.425–0.520)	0.404
RPPA	No adjustment	50	0.472 (0.425–0.520)	0.404
RPPA	No adjustment	100	0.472 (0.425–0.520)	0.404
RPPA	Shift (δ)	10	0.610 (0.526–0.695)	0.404
RPPA	Shift (δ)	20	0.614 (0.530–0.689)	0.404
RPPA	Shift (δ)	50	0.617 (0.530–0.692)	0.404
RPPA	Shift (δ)	100	0.618 (0.532–0.689)	0.404
RPPA	Shift + quantile	10	0.761 (0.583–0.904)	0.555
RPPA	Shift + quantile	20	0.792 (0.640–0.904)	0.585
RPPA	Shift + quantile	50	0.794 (0.670–0.876)	0.581
RPPA	Shift + quantile	100	0.800 (0.686–0.863)	0.585
ProCan	No adjustment	10	0.616 (0.591–0.637)	0.404
ProCan	No adjustment	20	0.616 (0.591–0.637)	0.404
ProCan	No adjustment	50	0.616 (0.591–0.637)	0.404
ProCan	No adjustment	100	0.616 (0.591–0.637)	0.404
ProCan	Shift (δ)	10	0.603 (0.479–0.645)	0.404
ProCan	Shift (δ)	20	0.615 (0.557–0.641)	0.404
ProCan	Shift (δ)	50	0.622 (0.581–0.645)	0.404
ProCan	Shift (δ)	100	0.619 (0.583–0.641)	0.404
ProCan	Shift + quantile	10	0.722 (0.551–0.904)	0.529
ProCan	Shift + quantile	20	0.778 (0.661–0.878)	0.578
ProCan	Shift + quantile	50	0.780 (0.682–0.858)	0.568
ProCan	Shift + quantile	100	0.795 (0.738–0.846)	0.590
APOLLO	No adjustment	10	0.824 (0.817–0.831)	0.404
APOLLO	No adjustment	20	0.824 (0.817–0.831)	0.404
APOLLO	No adjustment	50	0.824 (0.817–0.831)	0.404
APOLLO	No adjustment	100	0.824 (0.817–0.831)	0.404
APOLLO	Shift (δ)	10	0.807 (0.712–0.834)	0.404
APOLLO	Shift (δ)	20	0.817 (0.754–	0.404

Platform	Method	K	Mean 80% coverage (2.5th–97.5th)	Mean width
			0.834)	
APOLLO	Shift (δ)	50	0.826 (0.812–0.833)	0.404
APOLLO	Shift (δ)	100	0.827 (0.814–0.835)	0.404
APOLLO	Shift + quantile	10	0.705 (0.435–0.924)	0.341
APOLLO	Shift + quantile	20	0.765 (0.551–0.913)	0.369
APOLLO	Shift + quantile	50	0.775 (0.664–0.857)	0.358
APOLLO	Shift + quantile	100	0.789 (0.725–0.861)	0.367

Supplementary Table S7. openFDA and Drugs@FDA search for antibody–drug conjugates (search date 15 October 2026; approval cutoff 2 October 2026).

Queries and the number of records returned. A count of 0 is an empty response, not a failed request. The active-ingredient queries were added after openfda.generic_name:mafodotin returned 0 and missed Blenrep; the belantamab label query was added because application BLA761440 has no openFDA label record; the sunirine query was added for a phrase hit whose payload stem was not in the planned list.

Endpoint	Query	Hits	Note
label	"antibody-drug conjugate"	20	label text contains the phrase
drugsfda	openfda.generic_name:vedotin	5	planned generic-name stem query
drugsfda	openfda.generic_name:mafodotin	0	planned generic-name stem query
drugsfda	openfda.generic_name:deruxtecan	2	planned generic-name stem query
drugsfda	openfda.generic_name:govitecan	1	planned generic-name stem query
drugsfda	openfda.generic_name:emtansine	1	planned generic-name stem query
drugsfda	openfda.generic_name:soravtansine	1	planned generic-name stem query
drugsfda	openfda.generic_name:tesirine	1	planned generic-name stem query
drugsfda	openfda.generic_name:ozogamicin	2	planned generic-name stem query
drugsfda	openfda.generic_name:duocarmazine	0	planned generic-name stem query

Endpoint	Query	Hits	Note
drugsfda	openfda.generic_name:tirumotecan	0	planned generic-name stem query
drugsfda	products.active_ingredients.name:vedotin	6	added after openfda.generic_name:mafodotin returned 0 and missed Blenrep
drugsfda	products.active_ingredients.name:mafodotin	1	added after openfda.generic_name:mafodotin returned 0 and missed Blenrep
drugsfda	products.active_ingredients.name:deruxtecan	3	added after openfda.generic_name:mafodotin returned 0 and missed Blenrep
drugsfda	products.active_ingredients.name:govitecan	1	added after openfda.generic_name:mafodotin returned 0 and missed Blenrep
drugsfda	products.active_ingredients.name:emtansine	1	added after openfda.generic_name:mafodotin returned 0 and missed Blenrep
drugsfda	products.active_ingredients.name:soravtansine	1	added after openfda.generic_name:mafodotin returned 0 and missed Blenrep
drugsfda	products.active_ingredients.name:tesirine	1	added after openfda.generic_name:mafodotin returned 0 and missed Blenrep
drugsfda	products.active_ingredients.name:ozogamicin	2	added after openfda.generic_name:mafodotin returned 0 and missed Blenrep
drugsfda	products.active_ingredients.name:duocarmazine	0	added after openfda.generic_name:mafodotin returned 0 and missed Blenrep
drugsfda	products.active_ingredients.name:tirumotecan	0	added after openfda.generic_name:mafodotin returned 0 and missed Blenrep
label	"belantamab"	2	added because the Blenrep application number has no openFDA label record
drugsfda	products.active_ingredients.name:sunirine	1	phrase hit whose payload stem is not in the planned list

Products whose active-ingredient name contained a payload stem, plus the phrase hit whose mechanism says the product is an antibody–drug conjugate. The HGNC symbol is the symbol whose name appears in the quoted mechanism text. Two applications of Adcetris and two of Datroway are listed separately. Of the 29 atlas targets, this search found an approved product

for ERBB2, TACSTD2, NECTIN4, F3, FOLR1 and MET (eight products). For the other 23 targets, no product in this search acted on the gene.

Brand	Generic name	Application	First approval	Label revision	Mechanism of action (quoted)	HGNC symbol
ADCETRIS	BRENTUXIMAB VEDOTIN	BLA125388	2011-08-19	2025-11-11	CD30 is a member of the tumor necrosis factor receptor family and is expressed on the surface of sALCL cells and on Hodgkin Reed-Sternberg (HRS) cells in cHL. Brentuximab vedotin is an antibody-drug conjugate (ADC).	TNFRSF8
ADCETRIS	BRENTUXIMAB VEDOTIN	BLA125399	2011-08-19	2025-11-11	CD30 is a member of the tumor necrosis factor receptor family and is expressed on the surface of sALCL cells and on Hodgkin Reed-Sternberg (HRS) cells in cHL. Brentuximab vedotin is an antibody-drug conjugate (ADC). no label record for BLA125399; mechanism quoted from BLA125388, the other application	TNFRSF8

Brand	Generic name	Application	First approval	Label revision	Mechanism of action (quoted)	HGNC symbol
					of the same brand	
KADCYLA	ADO-TRASTUZUMAB EMTANSINE	BLA125427	2013-02-22	2026-04-01	Ado-trastuzumab emtansine is a HER2-targeted antibody-drug conjugate.	ERBB2
BESPONSA	INOTUZUMAB OZOGAMICIN	BLA761040	2017-08-17	2026-02-06	Inotuzumab ozogamicin is a CD22-directed antibody drug conjugate (ADC).	CD22
MYLOTARG	GEMTUZUMAB OZOGAMICIN	BLA761060	2017-09-01	2025-12-19	Gemtuzumab ozogamicin is a CD33-directed antibody-drug conjugate (ADC).	CD33
TRODELVY	SACITUZUMAB GOVITECAN-HZIY	BLA761115	2020-04-22	2026-06-24	Sacituzumab govitecan-hziy is a Trop-2-directed antibody-drug conjugate.	TACSTD2
POLIVY	POLATUZUMAB VEDOTIN-PIIQ	BLA761121	2019-06-10	2026-08-24	Polatuzumab vedotin-piiq is a CD79b-directed antibody-drug conjugate with activity against dividing B cells.	CD79B
PADCEV	ENFORTUMAB VEDOTIN-EJFV	BLA761137	2019-12-18	2026-08-04	The antibody is a human IgG1 kappa directed against Nectin-4, an adhesion protein located on the surface of cells. Enfortumab	NECTIN4

Brand	Generic name	Application	First approval	Label revision	Mechanism of action (quoted)	HGNC symbol
					vedotin-ejfv is an ADC.	
ENHERTU	FAM-TRASTUZUMAB DERUXTECAN-NXKI	BLA761139	2019-12-20	2026-05-15	Fam-trastuzumab deruxtecan-nxki is a HER2-directed antibody-drug conjugate.	ERBB2
ZYNLONTA	LONCASTUXIMAB TESIRINE-LPYL	BLA761196	2021-04-23	2026-06-12	Loncastuximab tesirine-lpyl is an antibody-drug conjugate (ADC) targeting CD19.	CD19
TIVDAK	TISOTUMAB VEDOTIN-TFTV	BLA761208	2021-09-20	2025-11-20	Tisotumab vedotin-tftv is a tissue factor (TF)-directed antibody drug conjugate (ADC).	F3
ELAHERE	MIRVETUXIMAB SORAVTANSINE-GYNX	BLA761310	2022-11-14	2025-07-14	The antibody is a chimeric IgG1 directed against folate receptor alpha (FRα). Mirvetuximab soravtansine-gynx is an antibody-drug conjugate (ADC).	FOLR1
EMRELIS	TELISOTUZUMAB VEDOTIN-TLLV	BLA761384	2025-05-14	2025-05-25	Telisotuzumab vedotin-tllv is a c-Met-directed antibody drug conjugate (ADC).	MET
DATROWAY	DATOPOTAMAB DERUXTECAN-DLNK	BLA761394	2025-01-17	2026-05-27	Datopotamab deruxtecan-dlnk, is a Trop-2-directed	TACSTD2

Brand	Generic name	Application	First approval	Label revision	Mechanism of action (quoted)	HGNC symbol
					antibody-drug conjugate.	
BLENREP	BELANTAMAB MAFODOTIN-BLMF	BLA761440	2025-10-23	2025-10-23	The antibody component is an afucosylated IgG1 directed against BCMA, a protein expressed on normal B lymphocytes and multiple myeloma cells. Belantamab mafodotin-blmf is an antibody-drug conjugate (ADC).	TNFRSF17
DECNUPAZ	PIVEKIMAB SUNIRINE-PVZY	BLA761460	2026-05-27	2026-05-27	Pivekimab sunirine-pvzy is a CD123 (alpha-subunit of the interleukin-3 receptor)-directed antibody-drug conjugate (ADC).	IL3RA
DATROWAY	DATOPOTAMAB DERUXTECAN-DLNK	BLA761464	2025-06-23	2026-05-27	Datopotamab deruxtecan-dlnk, is a Trop-2-directed antibody-drug conjugate. no label record for BLA761464; mechanism quoted from BLA761394, the other application of the same brand	TACSTD2

Supplementary Table S8. Pairs in each level of the cascade lookup.

Level	RPPA pairs (antibodies)	ProCan pairs (proteins)	APOLLO genes
L1, matched cancer	1,206 (127)	3,356 (2,091)	6,982
L2, CPTAC mean	2,653 (127)	2,616 (2,570)	93
L3, ProCan mean	62 (2)	not used	26
L4, global mean	124 (4)	146 (79)	95
Total	4,045 (134)	6,118 (2,746)	7,196

Supplementary Table S9. Downloaded files.

File	URL	Bytes	SHA-256	Downloaded
data/raw/cptac/BRCA/BRCA_meta.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/BRCA/BRCA_meta.txt	7369	16d00a3022fbafa9d29e74c0ee689ae63279911f04aeef2d21344ed51a0a5cb7	2026-10-01T06:03:52Z
data/raw/cptac/BRCA/BRCA_Tumor_CaseList.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/BRCA/BRCA_Tumor_CaseList.txt	1094	27c5bbd3b25f2421db1fcc174a1d0cde9835a191eb8b8d957241ef0b531b9993	2026-10-01T06:03:53Z
data/raw/cptac/BRCA/BRCA_phenotype.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/BRCA/BRCA_phenotype.txt	227024	7f9fa329e1352695ea898de6caecde43dfb45445c63911d9b649747517dd1091	2026-10-01T06:03:54Z
data/raw/cptac/BRCA/BRCA_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/BRCA/BRCA_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt	33078031	1d7a10c7ed2df611ec807e08e0f8e0694689a8582f9c3a98fbc34235b002d819	2026-10-01T06:04:12Z
data/raw/cptac/BRCA/BRCA_proteomics_gene_abundance_log2_reference_intensity_normalized	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/BRCA/BRCA_proteomics	21398351	e584331bc77a440af50692c7874b736b3d135d01d96ce95d66af18cc64009b21	2026-10-01T06:04:21Z

File	URL	Bytes	SHA-256	Downloaded
_Tumor.txt	_gene_abundance_log2_reference_intensity_normalized_Tumor.txt			
data/raw/cptac/CCRCC/CCRCC_meta.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/CCRCC/CCRCC_meta.txt	10979	c332231f3cbd204094236dfd302977e70bbdddcf0049e862b60dd10244d04dae	2026-10-01T06:04:22Z
data/raw/cptac/CCRCC/CCRCC_Tumor_CaseList.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/CCRCC/CCRCC_Tumor_CaseList.txt	1030	ee692071c563ae797e5d7670e2b8698049e2467eb30f5916f7707e798d1441be	2026-10-01T06:04:23Z
data/raw/cptac/CCRCC/CCRCC_Normal_CaseList.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/CCRCC/CCRCC_Normal_CaseList.txt	800	4e631863d653d09b11e249176f3318c96928240f02f631725b2053be7d92f6ea	2026-10-01T06:04:24Z
data/raw/cptac/CCRCC/CCRCC_phenotype.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/CCRCC/CCRCC_phenotype.txt	191249	929ce841da91caf2a03a58b1561929648f4be50b8b081f7182e2ad95ee936d0a	2026-10-01T06:04:25Z
data/raw/cptac/CCRCC/CCRCC_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/CCRCC/CCRCC_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt	33113844	bcef3edfb1fcc738859369d78f2b7edb3f0972d438b4b1f50a114519f2ab9a50	2026-10-01T06:05:17Z
data/raw/cptac/CCRCC/CCRCC_RNAse	https://cptac-pancancer-data.s3.us-west-2.amazonaws.	23810094	e4505de10482e5738937ebe7c9b4239ffdc86e5	2026-10-01T06:06:15Z

File	URL	Bytes	SHA-256	Downloaded
q_gene_RSEM_coding_UQ_1500_log2_Normal.txt	com/data_freeze_v1.2_reorganized/CCRCC/CCRCC_RNAseq_gene_RSEM_coding_UQ_1500_log2_Normal.txt		1cb092db7f2473e92040aa727	
data/raw/cptac/CCRCC/CCRCC_proteomics_gene_abundance_log2_reference_intensity_normalized_Tumor.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/CCRCC/CCRCC_proteomics_gene_abundance_log2_reference_intensity_normalized_Tumor.txt	16629110	090077c25f0eeb59b737912f39dec98901db949e1f143bbc481d5cd731550f60	2026-10-01T06:06:20Z
data/raw/cptac/CCRCC/CCRCC_proteomics_gene_abundance_log2_reference_intensity_normalized_Normal.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/CCRCC/CCRCC_proteomics_gene_abundance_log2_reference_intensity_normalized_Normal.txt	13026061	1b00d684cd99d870bee199128485c22d6cb079e0a01de3c044b7a4fdb1735618	2026-10-01T06:06:25Z
data/raw/cptac/COAD/COAD_meta.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/COAD/COAD_meta.txt	11672	5242103f57fb5d919ab7e228dd36c97b669915d6bee88ae1e5a43c3aa270295e	2026-10-01T06:06:26Z
data/raw/cptac/COAD/COAD_Tumor_CaseList.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/COAD/COAD_Tumor_CaseList.txt	988	1495c045402199d98670c8862ac01035b5521b3b8538b38c117fb80340009ac8	2026-10-01T06:06:27Z
data/raw/cptac/COAD/COAD_Normal_CaseList.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_r	898	7e43c6619464ae23a7c1162e9266f0c71175946fa290b37e0405a2e28a4f390c	2026-10-01T06:06:28Z

File	URL	Bytes	SHA-256	Downloaded
	eorganized/COAD/COAD_Normal_CaseList.txt			
data/raw/cptac/COAD/COAD_phenotype.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/COAD/COAD_phenotype.txt	204627	959ec27c2b668bfd2b0cd9ff811319efe62d3078c37c6ca590aa8be8b3b225d9	2026-10-01T06:06:29Z
data/raw/cptac/COAD/COAD_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/COAD/COAD_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt	28863817	e7c5108dcea2aa615b0c1288d7a8b43bdc1c2469f1a9543a4078f7ee048ef253	2026-10-01T06:06:37Z
data/raw/cptac/COAD/COAD_proteomics_gene_abundance_log2_reference_intensity_normalized_Tumor.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/COAD/COAD_proteomics_gene_abundance_log2_reference_intensity_normalized_Tumor.txt	12293004	fa2dde5f9165acdf6320d28e01dbad24b735ede320f5fab4a6ceab6c15aa0580	2026-10-01T06:06:42Z
data/raw/cptac/COAD/COAD_proteomics_gene_abundance_log2_reference_intensity_normalized_Normal.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/COAD/COAD_proteomics_gene_abundance_log2_reference_intensity_normalized_Normal.txt	12515956	a76a0cc0ab3986635e8745028eb0bcc64470035ab737cc0937ec7be114a09177	2026-10-01T06:06:47Z
data/raw/cptac/GBM/GBM_meta.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/GBM/	6809	6617ef2f9010acec1eba09a765ca3e35e4ba2ea7ba75ba9e8734ed9ad94feecc	2026-10-01T06:06:48Z

File	URL	Bytes	SHA-256	Downloaded
data/raw/cptac/GBM/GBM_Tumor_CaseList.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/GBM/GBM_Tumor_CaseList.txt	1087	51dc5cda8b5974690e9065751c5c3437b04ea210dfc895bcaee2e960e40ff2e1	2026-10-01T06:06:48Z
data/raw/cptac/GBM/GBM_phenotype.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/GBM/GBM_phenotype.txt	209875	807c6b83236dfb96c10b8ad224f68b21fb356bca4bd14514095cb1c5256ea575	2026-10-01T06:06:50Z
data/raw/cptac/GBM/GBM_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/GBM/GBM_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt	32926067	2c3c0e0e8f7f5466a3d078cf868e1f7f5e3529b5d7ee04eabec97e9ade9a1ea1	2026-10-01T06:06:58Z
data/raw/cptac/GBM/GBM_proteomics_gene_abundance_log2_reference_intensity_normalized_Tumor.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/GBM/GBM_proteomics_gene_abundance_log2_reference_intensity_normalized_Tumor.txt	18033471	d7393915eb8ec10c1c48bea03d78cd9ee2a3477f7bb4b93f808ffade03c2f01b	2026-10-01T06:07:06Z
data/raw/cptac/HNSCC/HNSCC_meta.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/HNSCC/HNSCC_meta.txt	12087	3c9e32ff8b51b4a22f967ca753185e01d79f69d860f9530f5587314bd06c80ac	2026-10-01T06:07:07Z
data/raw/cptac/HNSCC/HNSCC_Tumor_CaseList.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/HNSCC/	1080	18037cf6f7322421d06094aab41cff95ad1ea0d3cbf8298013a1d315feb3c4e4	2026-10-01T06:07:08Z

File	URL	Bytes	SHA-256	Downloaded
	HNSCC_Tumor_CaseList.txt			
data/raw/cptac/HNSCC/HNSCC_Normal_CaseList.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/HNSCC/HNSCC_Normal_CaseList.txt	620	7c943da5e8f6152a399433a2f166294a2de327d7d0e6693dd75896747f32cec1	2026-10-01T06:07:09Z
data/raw/cptac/HNSCC/HNSCC_phenotype.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/HNSCC/HNSCC_phenotype.txt	214773	579f51424bfbff5f9baf17553c95c12dfd94fc4a51b2f3562768d7c255adbd10	2026-10-01T06:07:10Z
data/raw/cptac/HNSCC/HNSCC_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/HNSCC/HNSCC_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt	35019254	a308c4410eb347e9036dff918935804981f5c67d7a2f4f9003c47d41de8cf195	2026-10-01T06:07:20Z
data/raw/cptac/HNSCC/HNSCC_RNAseq_gene_RSEM_coding_UQ_1500_log2_Normal.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/HNSCC/HNSCC_RNAseq_gene_RSEM_coding_UQ_1500_log2_Normal.txt	15523255	bb190f7493f83d693e8232f43ebf6a3e09b12ea23f84528fa04f66e56d2638db	2026-10-01T06:07:25Z
data/raw/cptac/HNSCC/HNSCC_proteomics_gene_abundance_log2_reference_intensity_normalized_Tumor.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/HNSCC/HNSCC_proteomics_gene_abundance_log2_reference_intensity_normalized_Tumor.txt	18184418	6d52dac50c55a4aa30e0aabaf3047a3a099875ea075b65d8f0bc83ac45a5c4b3	2026-10-01T06:07:31Z

File	URL	Bytes	SHA-256	Downloaded
data/raw/cptac/HNSCC/HNSCC_proteomics_gene_abundance_log2_reference_intensity_normalized_Normal.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/HNSCC/HNSCC_proteomics_gene_abundance_log2_reference_intensity_normalized_Normal.txt	10542735	0a067886f9b553d3de24d2dbab21e18037c44a50a52b9d60004d22dba3442adc	2026-10-01T06:07:36Z
data/raw/cptac/LSCC/LSCC_meta.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/LSCC/LSCC_meta.txt	12051	6c334bbd8c9c4acfc94709a7fea5fd0181efa0881cee8ce1793085a5281a9d99	2026-10-01T06:07:37Z
data/raw/cptac/LSCC/LSCC_Tumor_CaseList.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/LSCC/LSCC_Tumor_CaseList.txt	1186	8198fcc0e10e6bfaaabbb9886437b8e805af11e8f581fbff5ba63ec562c5c0eb	2026-10-01T06:07:38Z
data/raw/cptac/LSCC/LSCC_Normal_CaseList.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/LSCC/LSCC_Normal_CaseList.txt	1087	5fdf25da74eba7a2f1cfeb3b7424b1cf6cc45cfc8163d721785abf908f424a27	2026-10-01T06:07:39Z
data/raw/cptac/LSCC/LSCC_phenotype.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/LSCC/LSCC_phenotype.txt	212329	34f17a653dd5ebce0cdf1257c4498fe6579e314d18d435d333e4bcc4db22ef8d	2026-10-01T06:07:40Z
data/raw/cptac/LSCC/LSCC_RNAseq_gene_RSEM_coding_UQ_1500_	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/	35179517	0da58ae002ccc65cc83f13038f7f8439ad672898822e545e475f99540f5b7377	2026-10-01T06:07:48Z

File	URL	Bytes	SHA-256	Downloaded
log2_Tumor.txt	LSCC/LSCC_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt			
data/raw/cptac/LSCC/LSCC_RNAseq_gene_RSEM_coding_UQ_1500_log2_Normal.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/LSCC/LSCC_RNAseq_gene_RSEM_coding_UQ_1500_log2_Normal.txt	30624657	61822d27dda6ef9cf67743e8b9511f84620042d1e703fac8c5b25cfdc2a27f07	2026-10-01T06:07:54Z
data/raw/cptac/LSCC/LSCC_proteomics_gene_abundance_log2_reference_intensity_normalized_Tumor.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/LSCC/LSCC_proteomics_gene_abundance_log2_reference_intensity_normalized_Tumor.txt	20631271	fd9f9cc0aed40942fd33d750d1d207bcdf9ea231fcddeff3cc8ac8fdf3e1384d	2026-10-01T06:08:00Z
data/raw/cptac/LSCC/LSCC_proteomics_gene_abundance_log2_reference_intensity_normalized_Normal.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/LSCC/LSCC_proteomics_gene_abundance_log2_reference_intensity_normalized_Normal.txt	18920447	ea3f8e5a1e42f708af87f6cdfc86d3fa63a96bfdc4f6f3316ba54380ea579aaf	2026-10-01T06:08:05Z
data/raw/cptac/LUAD/LUAD_meta.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/LUAD/LUAD_meta.txt	12232	f9facf713af130ad097c56794f1fbc9107334e02ad9bf0df3958b40d68b6af10	2026-10-01T06:08:06Z
data/raw/cptac/LUAD/LUAD_Tumor_CaseList.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/	1092	92aba1a2d482feccb2f3f1873e147758674ee7472519340d3859e0c43f983ddf	2026-10-01T06:08:07Z

File	URL	Bytes	SHA-256	Downloaded
	LUAD/LUAD_Tumor_CaseList.txt			
data/raw/cptac/LUAD/LUAD_Normal_CaseList.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/LUAD/LUAD_Normal_CaseList.txt	1010	c6425a338baa30cdf33fcb617946ba6693d90100e729bbcc4ece7e9eea0ce623	2026-10-01T06:08:08Z
data/raw/cptac/LUAD/LUAD_phenotype.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/LUAD/LUAD_phenotype.txt	210137	9f777402f7d7d53856a5adfc73c8291136430ddfad85641584426921609b349a	2026-10-01T06:08:09Z
data/raw/cptac/LUAD/LUAD_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/LUAD/LUAD_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt	35757367	c2c5e97fd469591d7aec13bd28ba68aa19fe7ced161592f69924ecc388513e05	2026-10-01T06:08:19Z
data/raw/cptac/LUAD/LUAD_RNAseq_gene_RSEM_coding_UQ_1500_log2_Normal.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/LUAD/LUAD_RNAseq_gene_RSEM_coding_UQ_1500_log2_Normal.txt	32856789	a54d7023a79ccd5159c01f70dbc6cec472e02b0ed9142d052c816cee51220b70	2026-10-01T06:08:28Z
data/raw/cptac/LUAD/LUAD_proteomics_gene_abundance_log2_reference_intensity_normalized_Tumor.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/LUAD/LUAD_proteomics_gene_abundance_log2_reference_intensity_normalized_	20132337	746900f54072198481207ea13f8a7ba14dcabd52345bace0bef7438120c03845	2026-10-01T06:08:36Z

File	URL	Bytes	SHA-256	Downloaded
	Tumor.txt			
data/raw/cptac/LUAD/LUAD_proteomics_gene_abundance_log2_reference_intensity_normalized_Normal.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/LUAD/LUAD_proteomics_gene_abundance_log2_reference_intensity_normalized_Normal.txt	18491662	3cfc62b159276837dea644f457ebf6b212964320e0b3d8ac87168d82eb73747c	2026-10-01T06:08:43Z
data/raw/cptac/OV/OV_meta.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/OV/OV_meta.txt	6029	ab6b3bb70460c682492a98f7620cbbeda61baab6e4d8bc1e25d19c0580390123	2026-10-01T06:08:44Z
data/raw/cptac/OV/OV_Tumor_CaseList.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/OV/OV_Tumor_CaseList.txt	664	2c122ca238e35876f1166d7edef43ab87bd03b1c6baadcd97b1d9a9715d46312	2026-10-01T06:08:45Z
data/raw/cptac/OV/OV_Normal_CaseList.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/OV/OV_Normal_CaseList.txt	160	318a82c237b00e8f47b86a6cd90eae685aff01b8d02a83b49b355608e1c09d6e	2026-10-01T06:08:46Z
data/raw/cptac/OV/OV_phenotype.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/OV/OV_phenotype.txt	157772	58c75e054cb98a551e6a06e21784eb70f428da14779ace530c5549bef9352e8b	2026-10-01T06:08:47Z
data/raw/cptac/OV/OV_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/OV/OV_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt	22239335	5b25a3cbea0bb041f40fccb8b4120cf2b8135f16f102aacb1e4cb39923509d97	2026-10-01T06:08:53Z

File	URL	Bytes	SHA-256	Downloaded
data/raw/cptac/OV/OV_proteomics_gene_abundance_log2_reference_intensity_normalized_Tumor.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/OV/OV_proteomics_gene_abundance_log2_reference_intensity_normalized_Tumor.txt	12506387	274c22512ab70e2a07197ca6e51d268038c90e903bac823c1b27b7950af4ae6f	2026-10-01T06:08:57Z
data/raw/cptac/OV/OV_proteomics_gene_abundance_log2_reference_intensity_normalized_Normal.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/OV/OV_proteomics_gene_abundance_log2_reference_intensity_normalized_Normal.txt	3007610	6fc249b27c9284b598a94860c1e36f4fd45843f65b14c4ccdfbc0153725bb07a	2026-10-01T06:09:00Z
data/raw/cptac/PDAC/PDAC_meta.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/PDAC/PDAC_meta.txt	11358	ce53f0acf92cbbbf165c8c27425c27b7b6de301c2e95279eaf124c37a2316d33	2026-10-01T06:09:01Z
data/raw/cptac/PDAC/PDAC_Tumor_CaseList.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/PDAC/PDAC_Tumor_CaseList.txt	1155	089f7a166f4b90e888c367aa1582d6bc675b572c7207ee4d7670daa325b1e36b	2026-10-01T06:09:02Z
data/raw/cptac/PDAC/PDAC_Normal_CaseList.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/PDAC/PDAC_Normal_CaseList.txt	484	9f1dcb19bfb6a4630901edbc7b4e1ad77d70fb2d47eb6c8459620a524a3dd9c4	2026-10-01T06:09:03Z
data/raw/cptac/PDAC/PDAC_phenotype.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_r	206286	91c4511765e6c99e3e9b1a3711f31d313a1dee408072dafe375fcfc6f1af4a52	2026-10-01T06:09:04Z

File	URL	Bytes	SHA-256	Downloaded
	eorganized/PDAC/PDAC_phenotype.txt			
data/raw/cptac/PDAC/PDAC_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/PDAC/PDAC_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt	34095120	3f00449e7112f8d15478ceb0563df1c0ad633823756638fdca66bef4986b4bad	2026-10-01T06:09:58Z
data/raw/cptac/PDAC/PDAC_RNAseq_gene_RSEM_coding_UQ_1500_log2_Normal.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/PDAC/PDAC_RNAseq_gene_RSEM_coding_UQ_1500_log2_Normal.txt	5852445	14c9e9b708763678a9fed459a81dc017564aa38afad9e667ab714f58594a8774	2026-10-01T06:10:19Z
data/raw/cptac/PDAC/PDAC_proteomics_gene_abundance_log2_reference_intensity_normalized_Tumor.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/PDAC/PDAC_proteomics_gene_abundance_log2_reference_intensity_normalized_Tumor.txt	16800925	e1be4bed67bfdac6fff7b12487b8af9e0e2a0e93901039f5f9124e180f24f6bd	2026-10-01T06:11:42Z
data/raw/cptac/PDAC/PDAC_proteomics_gene_abundance_log2_reference_intensity_normalized_Normal.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/PDAC/PDAC_proteomics_gene_abundance_log2_reference_intensity_normalized_Normal.txt	7177097	beb1bc33ee52d10f0f7a6fa0bfc0de97916471c37d80c5cb655ee0d0a2452ce9	2026-10-01T06:12:05Z
data/raw/cptac/UCEC/UCEC_meta.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/	13972	4058a8420129dd37f4d9035e1d87beb70a800ce9cb42754f74	2026-10-01T06:12:06Z

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	data_freeze_v1.2_reorganized/UCEC/UCEC_meta.txt		e7ad3d35f3020b	
data/raw/cptac/UCEC/UCEC_Tumor_CaseList.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/UCEC/UCEC_Tumor_CaseList.txt	949	9f8db0ff8d82a54f9fc962f53ca4300107d08bc644b734912eccc3147a647545	2026-10-01T06:12:07Z
data/raw/cptac/UCEC/UCEC_Normal_CaseList.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/UCEC/UCEC_Normal_CaseList.txt	97	79ee61ccfa92e1e8bb1f17b94a7176a062aa40e1c0acb46294af99c7a84fce9e	2026-10-01T06:12:08Z
data/raw/cptac/UCEC/UCEC_phenotype.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/UCEC/UCEC_phenotype.txt	197288	89ddc8c4953184d32a592cf0240ba97658fc2bb78e7caa37ab90dd74dd5e87ad	2026-10-01T06:12:10Z
data/raw/cptac/UCEC/UCEC_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/UCEC/UCEC_RNAseq_gene_RSEM_coding_UQ_1500_log2_Tumor.txt	30753085	3d708a346477a51f2aa1fc80479cd60b1888644145b69b1082c31c6405770882	2026-10-01T06:14:39Z
data/raw/cptac/UCEC/UCEC_proteomics_gene_abundance_log2_reference_intensity_normalized_Tumor.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/UCEC/UCEC_proteomics_gene_abundance_log2_reference_intensity_normalized_Tumor.txt	16560392	aea5d956b3496530faf86052886de19e857e96294a2dbcb3adb9702457865d36	2026-10-01T06:15:52Z

File	URL	Bytes	SHA-256	Downloaded
data/raw/cptac/UCEC/UCEC_proteomics_gene_abundance_log2_reference_intensity_normalized_Normal.txt	https://cptac-pancancer-data.s3.us-west-2.amazonaws.com/data_freeze_v1.2_reorganized/UCEC/UCEC_proteomics_gene_abundance_log2_reference_intensity_normalized_Normal.txt	3354332	225df3e7827fbeec499deba3d773828935dbe722559730a902c66f9dabe1d229	2026-10-01T06:16:07Z
data/raw/hgnc/hgnc_complete_set.txt	https://storage.googleapis.com/public-download-files/hgnc/tsv/tsv/hgnc_complete_set.txt	16953545	91d0ad20c34f26fb12c8cce3f80a65a6fc694f3a508040b19f6bfa29ae0d889e	2026-10-01T06:16:19Z
data/raw/ccle/protein_quant_current_normalized.csv.gz	https://gygi.hms.harvard.edu/data/ccle/protein_quant_current_normalized.csv.gz	30916232	b72a9ff39b123c801249ea212844ac1f500c3f38fa0f399ec6ad08ee1be2a7edefe6f26a1109880c37e8bc98c1b24d4bf217dd4b3520d78a7b2a15efae24f538	2026-10-01T06:20:10Z
data/raw/depmap/download_files_nocaptcha.csv	https://depmap.org/portal/api/no-captcha/download/files	163720	b7a0c1385e6cef30132b56aff61f1261d11e3f490490b355c430d32ee0dbdcfa	2026-10-01T06:20:12Z
data/raw/depmap/Model.csv	https://ndownloader.figshare.com/files/51065297	645696	2a71dc94110efcc0221eae821bb93a9f03b54bea16f005818911a09d33383d56	2026-10-01T06:20:18Z
data/raw/depmap/OmicsExpressionProteinCodingGenesTPMLogp1.csv	https://ndownloader.figshare.com/files/51065489	506628654	5e7adf7e8217e5a02389c299dce1266e74c7a3b786b8c177e56d12402fe783ac	2026-10-01T07:36:23Z
data/raw/procan/ProCan-DepMapSanger_protein_matrix_8498_averaged.txt	https://ndownloader.figshare.com/files/34411172	94222540	68ff17c98d3f16	2026-10-01T07:36:23Z
data/raw/	https://	82593		2026-10-

File	URL	Bytes	SHA-256	Downloaded
procan/ProCan-DepMapSanger_mapping_file_averaged.txt	ndownloader.figshare.com/files/34411133		3992418f66e4ea776d484eb11475b4a8e1d079767eaa703b7e	01T07:36:23Z
data/raw/procan/figshare_README.md	https://ndownloader.figshare.com/files/36155439	3218	aaa5ca2539d67f12dcbbd893faff523e328cc69a994e81c2eeff8de0d93102ca	2026-10-01T07:36:23Z
data/raw/depmap/README_24Q4.txt	https://ndownloader.figshare.com/files/51065795	43103	f744a5cf5c112e01276e772679157aac8e85cdc4a8f6d3b86d808e3bfd3328912	2026-10-01T07:36:23Z
data/raw/tcga/TCGA-RPPA-pancan-clean.xena.gz	https://tcga-pancan-atlas-hub.s3.us-east-1.amazonaws.com/download/TCGA-RPPA-pancan-clean.xena.gz	5602638	2dca46e09280d2c26f9f83e82ee98c8c41d1b896f2966918529e913bfad1868d	2026-10-01T07:36:23Z
data/raw/tcga/EB++AdjustPANCAN_IlluminaHiSeq_RNASeqV2.geneExp.xena.gz	https://tcga-pancan-atlas-hub.s3.us-east-1.amazonaws.com/download/EB%2B%2BAdjustPANCAN_IlluminaHiSeq_RNASeqV2.geneExp.xena.gz	331000731	a00532ec86af8c07630c618f10f6277f09c484d0a9c17db5901edf95c7714b38	2026-10-01T07:36:24Z
data/raw/tcga/TCGA_phenotype_denseDataOnlyDownload.tsv.gz	https://tcga-pancan-atlas-hub.s3.us-east-1.amazonaws.com/download/TCGA_phenotype_denseDataOnlyDownload.tsv.gz	61165	ac3923bb155ada3391b653a811f6c8c9c4734593b0c2b0a1555728928d83e5a2	2026-10-01T07:36:24Z
data/raw/hpa/normal_ihc_data.tsv.zip	https://www.proteinatlas.org/download/tsv/normal_ihc_data.tsv.zip	5732831	fa2315e5a53ecffd9427a5bfe219c8cc41b54089ef6f6be7b5d79b5455e0677f	2026-10-01T07:36:24Z
data/raw/hpa/cancer_data.tsv.zip	https://www.proteinatlas.org/download/tsv/cancer_data.tsv.zip	1724018	6e771693dc1019a3536d8834175ee97dcfde4f835d5cb3bed7c5a48803668135	2026-10-01T07:36:24Z
data/raw/	https://	957555	419f65d1d6534	2026-10-

File	URL	Bytes	SHA-256	Downloaded
cmp/model_list_20260921.csv	cog.sanger.ac.uk/cmp/download/model_list_20260921.csv		454194f1a744550fab632cd1d7411f6b13411a885735aa37006	01T07:36:24Z
data/raw/docs/sanger_expression.html	https://depmap.sanger.ac.uk/documentation/datasets/expression/	30458	ff89f0a5bb3bef4e588e6cac4a1433229a807c236390bdc4eb12d0c7012686bc	2026-10-01T07:36:24Z
data/raw/docs/sanger_proteomics.html	https://depmap.sanger.ac.uk/documentation/datasets/proteomics/	27484	a55e5ba6405f4cd92b61d5abc6252c49b38bb68fc4a85096d65b9f8e781c1c21	2026-10-01T07:36:24Z
data/raw/cmp/rnaseq_merged_20260323.zip	https://cog.sanger.ac.uk/cmp/download/rnaseq_merged_20260323.zip	1291769946	d2a6acd70647cc4098a990cdedaca564c74411a87385b6f0d33e8081ce27b4c9	2026-10-01T08:03:27Z
data/raw/rppa_map/acc_tcga_pan_can_atlas_2018.antibody_col1.txt	https://media.githubusercontent.com/media/cBioPortal/datahub/master/public/acc_tcga_pan_can_atlas_2018/data_rppa.txt	2953	5160f8582dc0e8d549ec1fce53c94b9e075bc5f8e0588114c309c19bab905773	2026-10-01T11:08:10Z
data/raw/rppa_map/blca_tcga_pan_can_atlas_2018.antibody_col1.txt	https://media.githubusercontent.com/media/cBioPortal/datahub/master/public/blca_tcga_pan_can_atlas_2018/data_rppa.txt	2953	5160f8582dc0e8d549ec1fce53c94b9e075bc5f8e0588114c309c19bab905773	2026-10-01T11:08:10Z
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data/raw/	https://media.githubuserco	2953	5160f8582dc0e	2026-10-01T11:08:10Z

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rppa_map/ kich_tcga_pan_ can_atlas_2018 .antibody_col1. txt	ntent.com/media/ cBioPortal/ datahub/master/ public/ kich_tcga_pan_can _atlas_2018/ data_rppa.txt		8d549ec1fce53 c94b9e075bc5f 8e0588114c309 c19bab905773	
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data/raw/rppa_map/ov_tcga_pan_can_atlas_2018.antibody_col1.txt	https://media.githubusercontent.com/media/cBioPortal/datahub/master/public/ov_tcga_pan_can_atlas_2018/data_rppa.txt	2953	5160f8582dc0e8d549ec1fce53c94b9e075bc5f8e0588114c309c19bab905773	2026-10-01T11:08:10Z
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data/raw/rppa_map/pcpg_tcga_pan_can_atlas_2018.antibody_col1.txt	https://media.githubusercontent.com/media/cBioPortal/datahub/master/public/pcpg_tcga_pan_can_atlas_2018/data_rppa.txt	2953	5160f8582dc0e8d549ec1fce53c94b9e075bc5f8e0588114c309c19bab905773	2026-10-01T11:08:10Z
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skcm_tcga_pan_can_atlas_2018.antibody_col1.txt	cBioPortal/datahub/master/public/skcm_tcga_pan_can_atlas_2018/data_rppa.txt		c94b9e075bc5f8e0588114c309c19bab905773	
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data/raw/rppa_map/thym_tcga_pan_can_atlas_2018.antibody_col1.txt	https://media.githubusercontent.com/media/cBioPortal/datahub/master/public/thym_tcga_pan_can_atlas_2018/data_rppa.txt	2953	5160f8582dc0e8d549ec1fce53c94b9e075bc5f8e0588114c309c19bab905773	2026-10-01T11:08:10Z
data/raw/rppa_map/ucec_tcga_pan_can_atlas_2018.antibody_col1.txt	https://media.githubusercontent.com/media/cBioPortal/datahub/master/public/ucec_tcga_pan_can_atlas_2018/data_rppa.txt	2953	5160f8582dc0e8d549ec1fce53c94b9e075bc5f8e0588114c309c19bab905773	2026-10-01T11:08:10Z
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data/raw/rppa_map/uvm_tcga_pan_can_atlas_2018.antibody_col1.txt	data_rppa.txt https://media.githubusercontent.com/media/cBioPortal/datahub/master/public/uvm_tcga_pan_can_atlas_2018/data_rppa.txt	2953	5160f8582dc0e8d549ec1fce53c94b9e075bc5f8e0588114c309c19bab905773	2026-10-01T11:08:10Z

Supplementary Table S10. Context map of the matched lookup (L1).

Evaluated data set	Context	Matched CPTAC cohort
RPPA	breast invasive carcinoma	BRCA
RPPA	kidney clear cell carcinoma	CCRCC
RPPA	colon adenocarcinoma; rectum adenocarcinoma	COAD
RPPA	glioblastoma multiforme	GBM
RPPA	head and neck squamous cell carcinoma	HNSCC
RPPA	lung adenocarcinoma	LUAD
RPPA	lung squamous cell carcinoma	LSCC
RPPA	ovarian serous cystadenocarcinoma	OV
RPPA	pancreatic adenocarcinoma	PDAC
RPPA	uterine corpus endometrioid carcinoma	UCEC
RPPA	other cancer types	none
ProCan	Large Intestine	COAD
ProCan	Lung	LUAD and LSCC (mean of finite values)
ProCan	Haematopoietic and Lymphoid	none
APOLLO	lung adenocarcinoma	LUAD

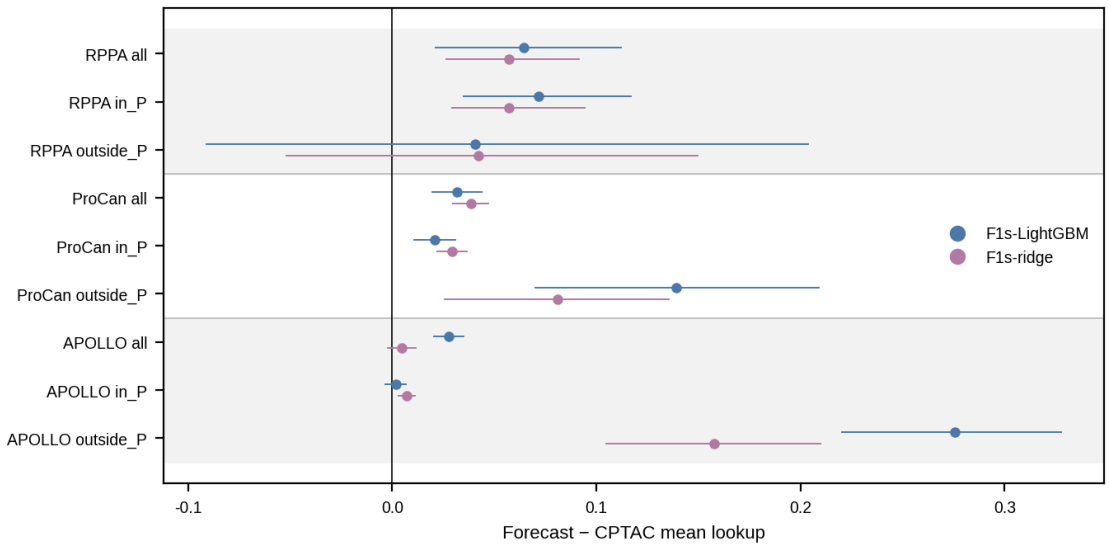
Supplementary figures

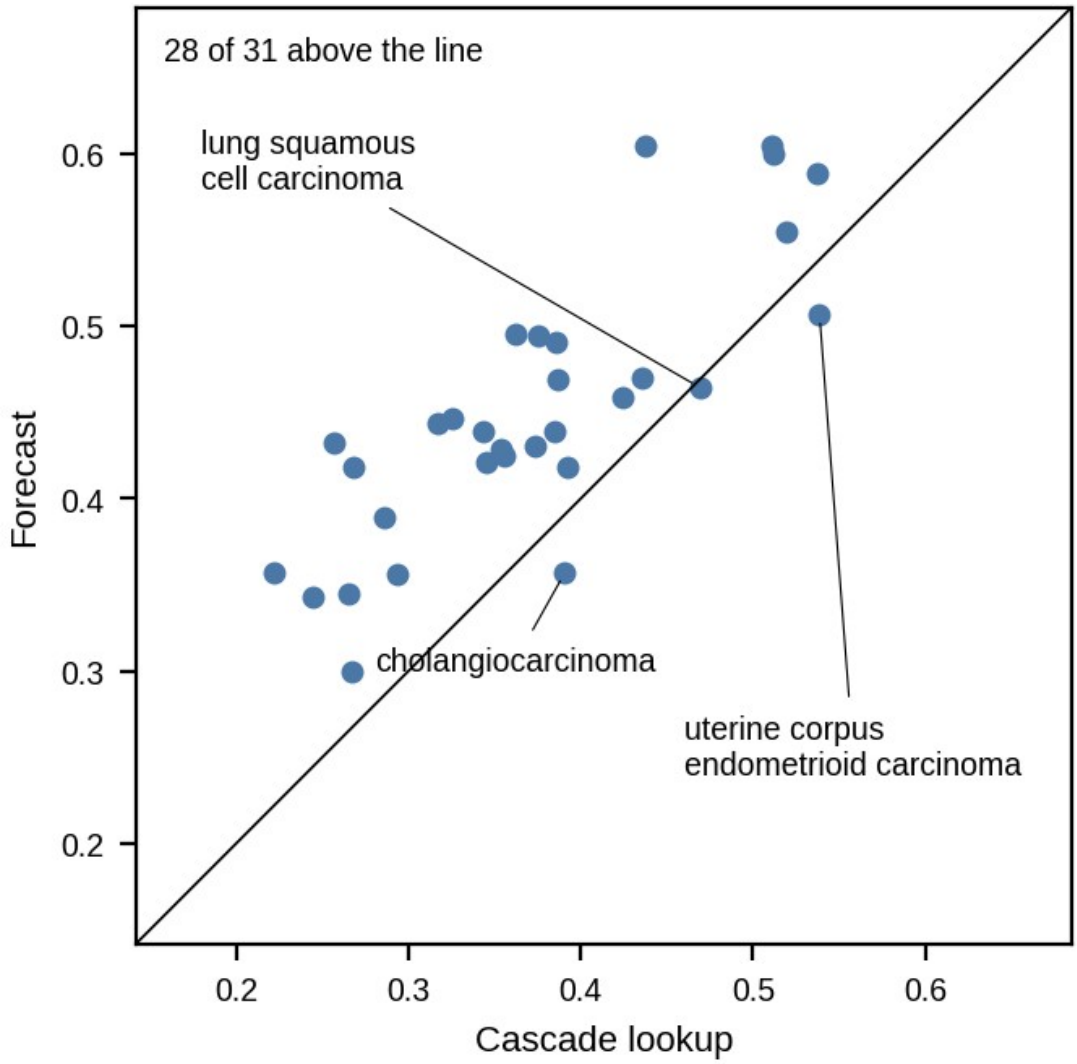
Supplementary Figure S1. Forecast minus CPTAC mean lookup by membership in the P gene set (post hoc). Differences in Spearman correlation with 95% intervals for F1s-LightGBM and F1s-ridge on all pairs, pairs inside P and pairs outside P, for RPPA, ProCan and APOLLO. Pale bands and horizontal lines separate the three platforms. Values are in Supplementary Table S2.

Supplementary Figure S2. Within-cancer-type comparison on RPPA (post hoc). Spearman correlation of the forecast against that of the cascade lookup with observed concordance in each of the 31 RPPA cancer types; the line marks equality. Cancer types below the line are labelled.

Supplementary references

[Converted from citation keys; numbered separately from the main text, or merged into the main reference list if the journal prefers. CHECK the Scientific Reports rule and follow it.]


RPPA all
RPPA in_P
RPPA outside_P
ProCan all
ProCan in_P
ProCan outside_P
APOLLO all
APOLLO in_P
APOLLO outside_P
F1s-LightGBM
F1s-ridge
-0.1
0.0
0.1
0.2
0.3
Forecast − CPTAC mean lookup


28 of 31 above the line
lung squamous
cell carcinoma
cholangiocarcinoma
uterine corpus
endometrioid carcinoma
Forecast
Cascade lookup
0.2
0.3
0.4
0.5
0.6